
2025 ANNUAL AUDIT REPORT ON THE SOUTHEAST ENERGY EXCHANGE MARKET

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I. OVERVIEW

This is the 2025 Annual Audit Report on the Southeast Energy Exchange Market (SEEM). This report covers activity in 2025, includes some historical data from prior years, and incorporates observations from 2026. SEEM is a regional energy market that centralizes the collection of bids and offers among multiple parties to create matches for bilateral trades every 15 minutes. SEEM began operating in November 2022 and currently has 24 members.¹

Trading Highlights:

- Trading volumes increased in 2025 to 1,303,000 MWh from 1,055,000 MWh in 2024 and from 704,000 in 2023.
- The monthly average volume of cleared trades was 109,000 MWh (compared to 88,000 MWh in 2024).
- The highest monthly trade volume in 2025 was 132,878 MWh in May, which was an all-time high.
- SEEM volumes indicate a distinct upward trend over time.
- Total production cost savings since market opening exceed \$30 million.

According to the Auditing Process under the SEEM Market Rules (Section VI.D of the rules) our auditing duties fall into two general areas:

- (1) analyzing and affirming SEEM inputs, constraints, and matching results; and
- (2) responding to certain inquiries and reporting on the SEEM operation and outcomes.

Our Annual Report focuses on the results of our auditing activity in 2025. We focus on market outcomes using the 2025 data as well as historical data prior to 2025 and data for part of 2026. While our report focuses on SEEM outcomes, we monitor SEEM continuously using data screens to verify the underlying modeling operated by Hartigen in accordance with specific market rules. Hence, we evaluate bids and offers, evaluate matches, and verify SEEM constraints based on specific requirements in the Auditing Process in the SEEM Market Rules. We report these results each month and have found the SEEM market operated in compliance with the SEEM rules.

¹ The initial 18 members are: Alabama Power Company; Georgia Power Company; Mississippi Power Company; Associated Electric Cooperative, Inc.; Dalton Utilities; Dominion Energy South Carolina, Inc.; Duke Energy Carolinas, LLC; Duke Energy Progress, LLC; Louisville Gas & Electric Company and Kentucky Utilities Company; North Carolina Municipal Power Agency Number 1; PowerSouth Energy Cooperative; North Carolina Electric Membership Corporation; Tennessee Valley Authority; Georgia System Operations Corporation; Georgia Transmission Corporation; Municipal Electric Authority of Georgia; Oglethorpe Power Corporation; and South Carolina Public Service Authority. The Florida member joining in June 2023 are: Seminole Electric Cooperative; Tampa Electric Company; Duke Energy Florida; Florida Power Corporation; TEA Gainesville Regional Utilities; and TEA JEA.

In evaluating the market outcomes, we review the SEEM performance from an overall market perspective, evaluating both trends in trading and use of transmission service. Market outcomes are consistent with increasing trading volumes and liquidity.

In Section IV, we present our conclusions and our recommendation to consider potential long-term expansion of the SEEM market.

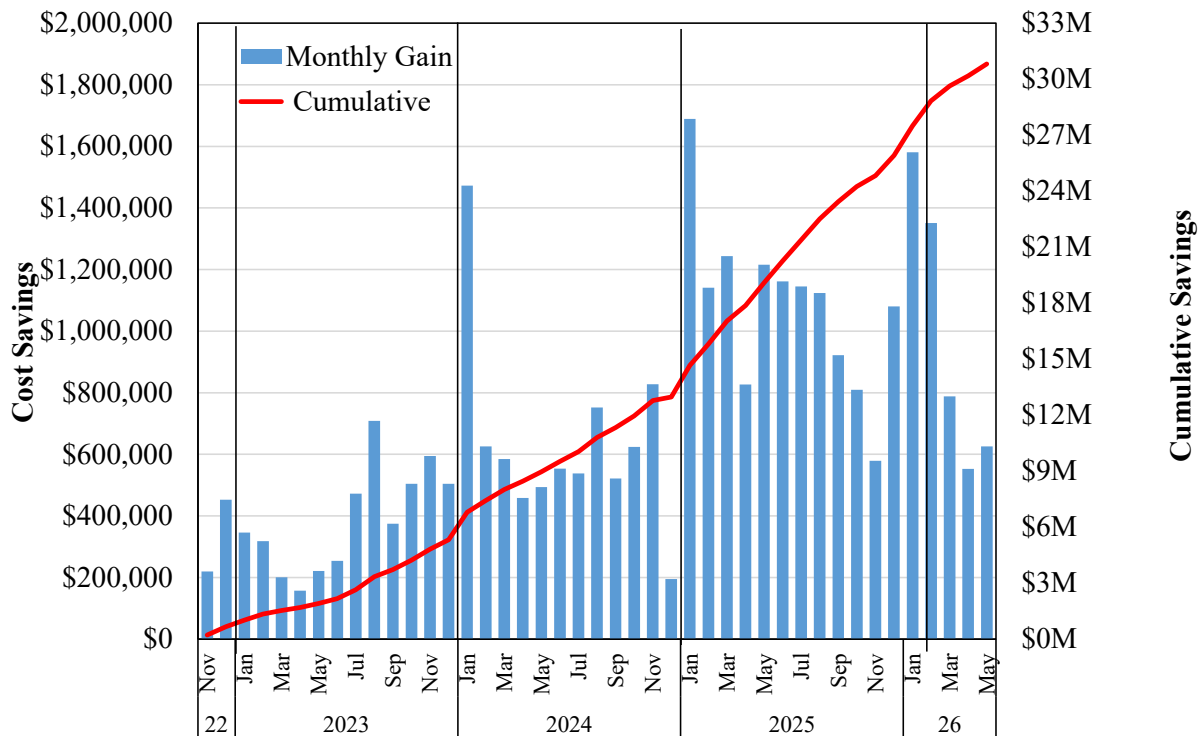
II. MARKET ACTIVITY

In this section, we review SEEM operations and outcomes in 2025 to discuss potential operating or market issues. This includes an overall review of the market trading, including volumes, prices, and characteristics of participation, and an evaluation of network usage, focusing on the key transmission paths and constraints.

A. Financial Results

Trades clearing in SEEM allow participants to reduce output from higher-cost resources and replace it with lower-cost ones, resulting in production cost savings. Cumulative production cost savings since market inception are at least \$30 million, and likely significantly more.² Figure 1 shows the (lower bound of) estimated production cost savings for each month since SEEM inception, including months in 2026. The red line shows the cumulative savings

Figure 1: Monthly Production Cost Savings

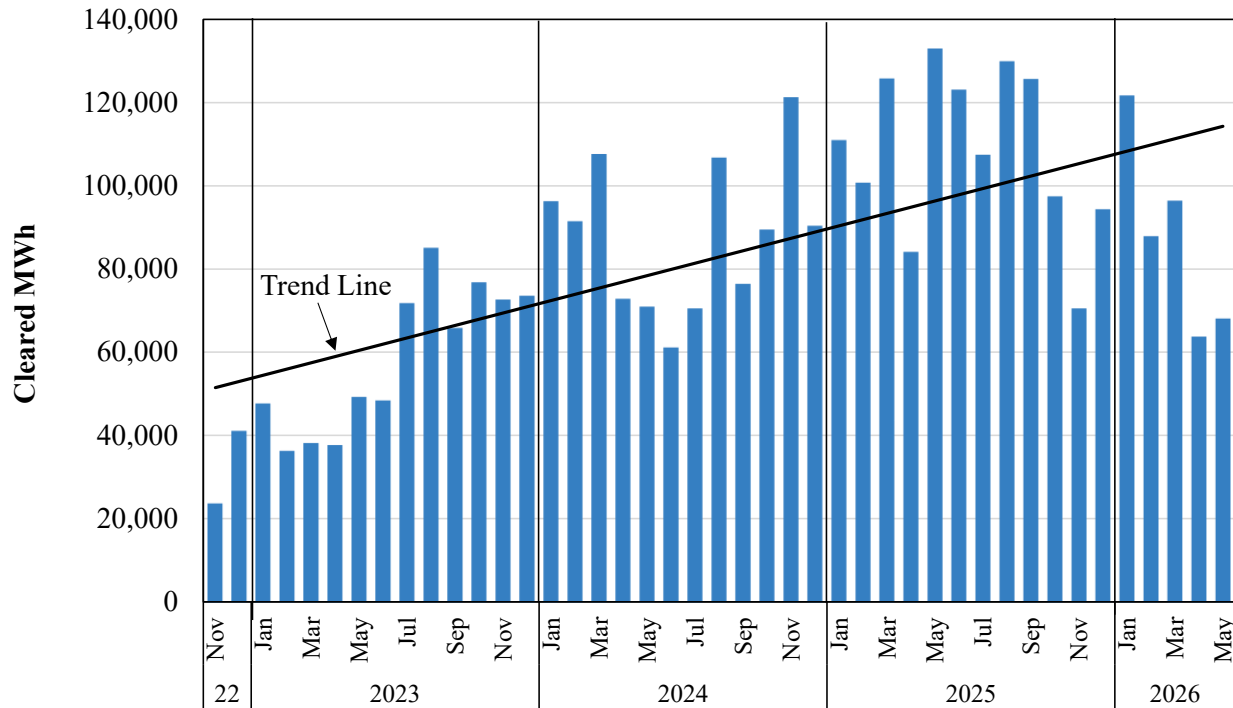


² There is likely significantly more production cost saving than the data shown because the bids (offers) are likely to be slightly lower than the true cost of buyers (higher than the true cost to sellers) due to the split-the-savings nature of SEEM. In a split-the-savings auction like SEEM, participants will improve their payoff by slightly lowering bids below actual avoided costs and slightly raising offers above production costs in an attempt to get a split closer to their counterparty's bid or offer.

B. Trading and Liquidity

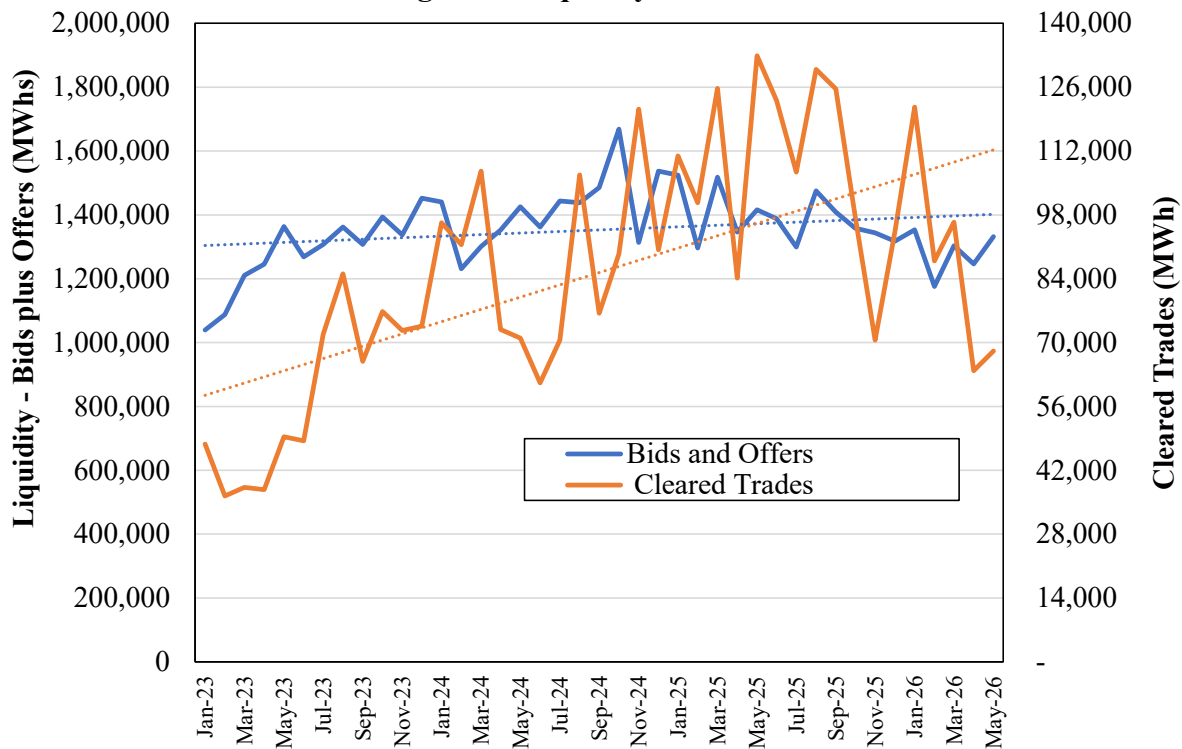
Trading has increased since the market opening in November 2022 as shown in Figure 2. This indicates increased interest and sustained confidence in the market.

Figure 2: Volume of Cleared Trades



Using total bids and offers as a measure of liquidity, Figure 3 compares liquidity to cleared trades to assess market development. The figure shows liquidity (bids and offers) is relatively steady, while cleared trades show a definite uptrend. This suggests offers and bids are converging to levels more likely to clear, which is positive for market performance.

Figure 3: Liquidity and Trades



We track the time series volumes of trades, offers, bids, and regional demand data (measured by Degree Days (DD))³ to gain insight into the performance of SEEM. We calculate statistics for the key data to identify key correlations.⁴ We found excess bids (bids less offers) are positively correlated with Degree Days, meaning higher regional load (using DD as the proxy) typically results in buyers turning to SEEM in higher volumes than sellers, likely due to sellers shoring up their own integrated operations under constrained conditions. This is what we have observed anecdotally as well.

Interestingly, cleared volume does not statistically correlate with DD, even as buyers tend to push into the market at these times. They appear to be chasing vanishing supply at these times. Indeed, this is consistent with the SEEM clearing price being statistically higher when DD increases (buyers are willing to pay more).

³ Degree days are the difference between the daily temperature mean and 65°F. If the temperature mean is above 65°F, we subtract 65 from the mean and the result is **Cooling Degree Days**. If the temperature mean is below 65°F, we subtract the mean from 65 and the result is **Heating Degree Days**. We use the absolute value of DD to reflect the

⁴ We use a correlation coefficient, which is a statistic that measures the relationship between two variables (in our case the cleared volumes and Degree Days). A positive correlation coefficient indicates the variables tend to move in the same direction while a negative correlation coefficient indicates the variables tend to move in opposite directions. A correlation coefficient at or close to zero means there is no linear relationship.

Next, we present data to provide insight into the extent to which losses and constraints have prevented economic transactions from clearing. Figure 4 shows the disposition of all economic bids and offers (i.e., those for which an economic counterparty was available). The blue bars show the economic bids and offers that were matched and cleared, while the transparent bars show those that did not clear because of the costs of losses, transmission constraints, or other constraints. The figure shows bid-offer pairs in tranches by the price spread. This is important because it allows us to infer why transactions likely did not clear.

To understand why economic bids and offers may not have cleared, we examine the bid-offer spread. Average loss charges are roughly \$2 per MWh, although some potential economic matches would incur higher loss costs and some less. Therefore, in the inset table, we divide totals between bid-offer spreads above and below \$3 per MWh. Those below \$3 are very likely not to have cleared because of the costs of losses, while most of those that did not clear at spreads above \$3 likely did not clear because of transmission constraints or participant constraints.

Figure 4: Cleared and Uncleared Economic Bids and Offers
2026

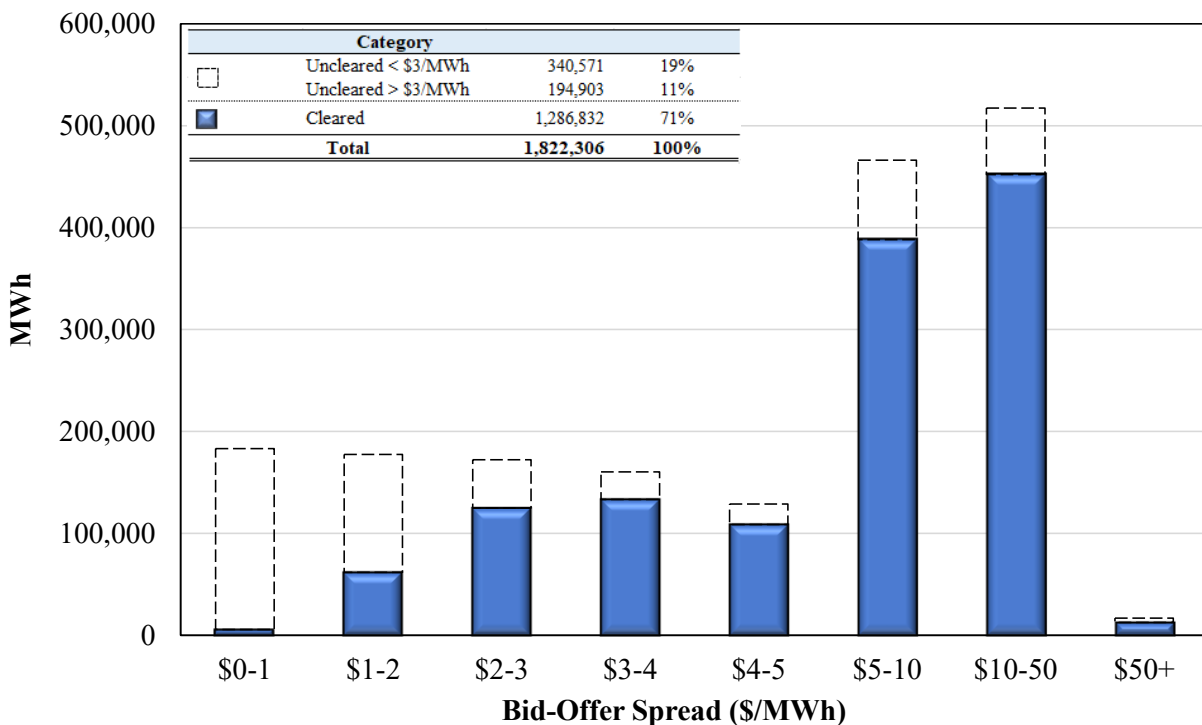


Figure 4 shows that over the entire period roughly 71 percent of the economic transactions cleared. It also shows that the costs of transmission losses were likely the most significant factor that prevented transactions from clearing, because most of the uncleared economic transactions were those with spreads less than \$3 per MWh.

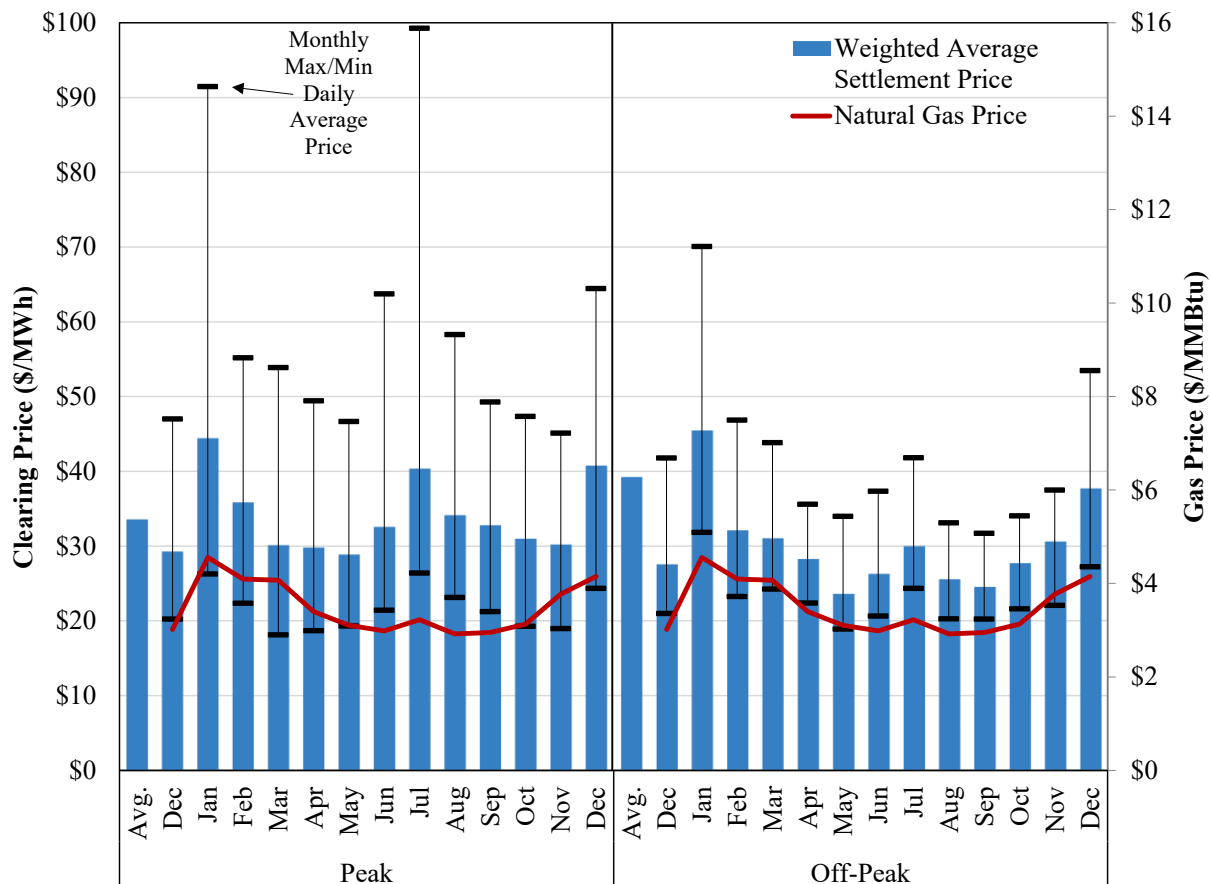
C. SEEM Prices

In this subsection, we report on pricing in the SEEM market. Figure 5 shows 2025 monthly SEEM clearing prices, natural gas costs, and average daily minimum and maximum prices in peak and off-peak hours during the month.

The figure shows that prices are correlated with natural gas costs, which is the marginal fuel in most hours. This supports the conclusion that the SEEM market has generally performed competitively since generators in a competitive market will face incentives to offer power at their marginal costs of producing it.

The superimposed “whisker lines” over the bars show the range of daily average prices in each month. The top of the lines is the highest daily average price during the month. The variation in prices throughout SEEM, which influences the variation in daily prices, is strongly influenced by the locational value of available resources.

Figure 5: Monthly Clearing Prices and Natural Gas Costs



Indeed, we calculated the weighted average price on all segments used in SEEM and we found a wide variance in the prices for transactions clearing over different segments. Ignoring segments that had no trades, the average price was about \$32/MWh for all segments. However, the average

prices by segment ranged from \$19/MWh to \$91/MWh. This shows that the value of transactions can vary significantly by path, mainly because transmission constraints can contribute to higher prices at different locations. If a constraint prevents higher total flows (economic trades) between two areas, the average price will be higher than if sufficient transmission capability were available to allow all economic trades to clear between the areas.

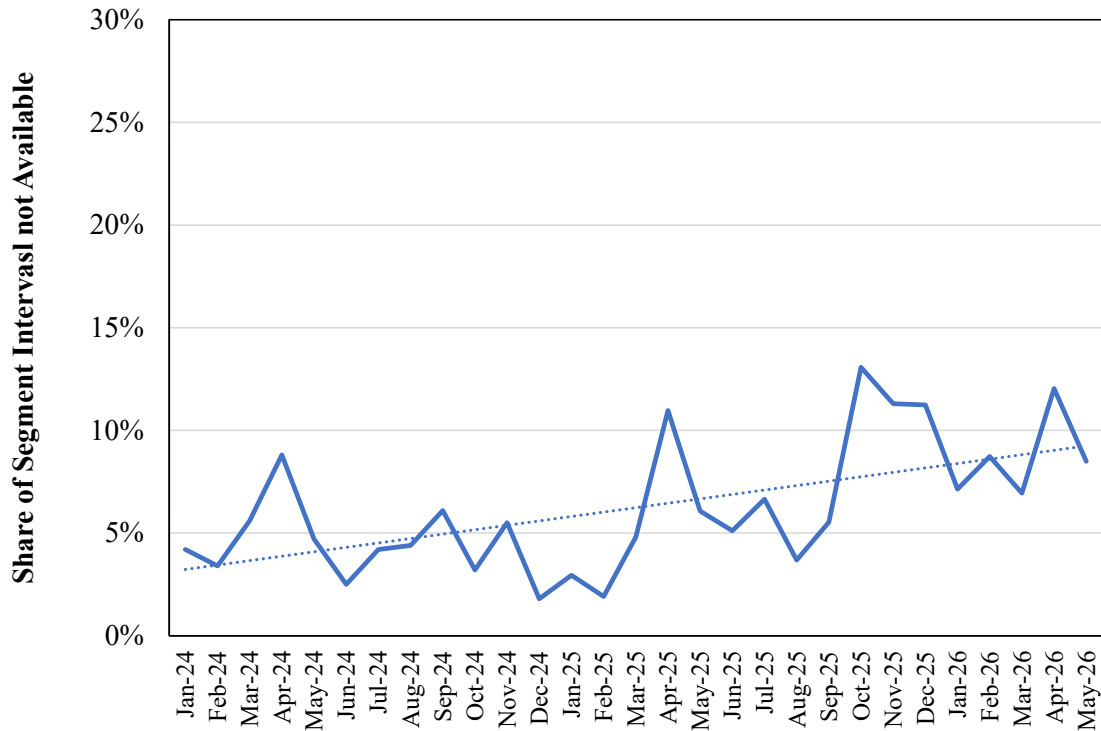
D. Network Usage

The SEEM model uses transmission segments made up of paths between directly adjacent systems. These segments are used by themselves or combined with connecting segments to form larger paths for trading. There are 359 segments. Most segments have an Available Transmission Capability (“ATC”) value that determines how much power can be transferred over the segment; others have no ATC value and are thus available without limit. Each month we evaluate the ATC and trading statistics for all SEEM segments available to the model.⁵

Our evaluation of segment usage has found that there are instances when the ATC is posted as zero or it is completely used by SEEM transactions. In these cases, SEEM is constrained and cannot clear additional trades using that segment. However, this happens in typically less than 10 percent of the time. Figure 6 shows the average monthly percentage of unavailable SEEM segments. Unavailability peaks in the shoulder seasons (spring and fall) when transmission maintenance is undertaken, which can result in reduced ATC values. While availability is high overall, we will monitor a recent trend showing increasing unavailability.

⁵ A segment is a fifteen-minute interval. We evaluate the ATC for all segments and all intervals (over one million per month).

Figure 6: Trend in Unavailable SEEM Segments



III. CONCLUSIONS AND RECOMMENDATIONS

Our evaluation of SEEM operations for 2025 has shown that implementation and operation of the SEEM platform continues to be successful.

Given the solid performance of the SEEM platform system, it holds promise to expand the system to include longer timeframes. The current SEEM market facilitates very short, near-term exchanges of economic energy. Additional benefits and efficiency could be achieved by transacting further ahead and for longer periods.

The main difference between these timeframes and the current market is that holders of transmission reservations would have to release the transmission. The current SEEM market uses unscheduled available transmission capability after the hourly schedules are processed. Hence, the transmission is released and made available at no cost. This would not be possible for longer lead-time transactions because the transmission reservations could potentially still have significant value in these timeframes. Therefore, implementing hourly real-time or day-ahead transactions would require:

- Offers by the holders of transmission reservations to make them available at a submitted cost (does not exist in the current market); and
- Bids by buyers and offers by sellers that are comparable to the current market.

Both of these recommended expansions would facilitate fuller utilization of the transmission system. However, an hourly day-ahead market would likely be the most beneficial because it could lead to changes in resource commitments that would lower the cost of serving load in the region.

As an illustration of potential benefits, we compare activity in SEEM and the bilateral hourly market to assess relative volumes. We make this comparison on a path basis by comparing volume on a SEEM path with the corresponding volume of hourly bilateral trades. We have all SEEM trade volumes needed for this comparison. For bilateral hourly trade volumes, we use transmission reservation (TSR) data from the Open Access Same-Time Information System (OASIS). Specifically, we use hourly point-to-point transmission service request data as a proxy for bilateral trades. Point-to-point TSRs are defined only on paths between adjacent transmission systems, so TSR data for hourly bilateral trades is available only on one-segment paths. These paths connect adjacent systems, and we use them as a proxy for bilateral trades between the two systems. Participants in the hourly bilateral market can string together multiple transmission reservations to create longer paths across multiple systems. However, because transmission costs exceed \$10/MWh, we conclude that this is a rare use of point-to-point reservations. Accordingly, we confine our comparison of SEEM volume and hourly bilateral volume to one-segment paths.

We collected data for April 2025 to April 2026. The data show slightly lower volume in SEEM than in the hourly bilateral market, indicating that most short-horizon economy energy for the period was traded in the hourly market. However, the monthly data do not always show this pattern. Table 1 compares volume over this time period. It shows higher total and average volume for OASIS, although the difference from SEEM volume is not significant and does not occur in every month.

Table 1: Path Volume Monthly Comparison

Top 20 SEEM Paths (MWh)		
Month	SEEM Volume	OASIS Hourly Reservations
Apr-25	33,329	109,217
May-25	68,120	125,779
Jun-25	59,215	64,920
Jul-25	58,423	16,286
Aug-25	65,488	36,988
Sep-25	70,331	48,353
Oct-25	41,637	12,247
Nov-25	29,686	10,406
Dec-25	39,906	34,110
Jan-26	50,935	101,875
Feb-26	35,024	73,578
Mar-26	39,248	16,521
Apr-26	33,721	10,943
Total	625,062	661,223
Average	48,082	50,863

The data suggests that, at least for the one-segment paths that facilitate trade between adjacent systems, SEEM markets have slightly lower volumes than the hourly bilateral markets. Expanding SEEM to the hourly market could bring these one-segment transactions into a multilateral mechanism and improve market access for buyers and sellers. More importantly, however, is the potential to extend hourly trades to multi-segment paths. SEEM clears trades very frequently on multi-segment paths; typically about 40 percent of trades used them. These trades take advantage of low-cost transmission across the footprint and pay only transmission losses. This is a substantial advantage over the hourly market, where transmission charges are pancaked and participants face accumulating transmission charges that are much higher -- over \$10/MWh per system compared to about \$2/MWh for SEEM. A multilateral mechanism that allows multi-segment paths to trade hourly with discounted transmission, as in SEEM, would further expand competitive offerings and increase production cost benefits for hourly activity.